

Public Economics and Policy

Lecture #7

Economic Analysis of Taxation (Part II)

September 15, 2026

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Learning objective

- ☐ Explain tax policies aimed at correcting negative externalities with a focus on Pigouvian tax principles.

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Outline

- ☐ Discussion on the assigned readings.
- ☐ Tax policy to correct for the negative externalities --- A Pigouvian tax
- ☐ Conclusion

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Assigned Readings

Everyone is expected to share their thoughts on all papers.

[1] Hattersley, L., & Mandeville, K. L. (2023). Global coverage and design of sugar-sweetened beverage taxes. *JAMA Network Open*, 6(3), e231412.

<https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2802843>

[2] Andreyeva, T., Marple, K., Marinello, S., Moore, T. E., & Powell, L. M. (2022). Outcomes following taxation of sugar-sweetened beverages: a systematic review and meta-analysis. *JAMA Network Open*, 5(6), e2215276.

<https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2792842>

[3] Chaloupka, F. J., Powell, L. M., & Warner, K. E. (2019). The use of excise taxes to reduce tobacco, alcohol, and sugary beverage consumption. *Annual review of public health*, 40(1), 187-201.

<https://www.annualreviews.org/content/journals/10.1146/annurev-publhealth-040218-043816>

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Reading

- ❑ Stiglitz, J. E., & Rosengard, J. K. (2015). *Economics of the public sector: Fourth international student edition*. WW Norton & Company. (Chapter 18)

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Further Discussion on Tax Evasion

Tax Evasion and the Shadow Economy

- ❑ Tax evasion as free riding (applies to all taxes, especially income tax and VAT)
- ❑ Expected value penalty = probability of detection x penalty level
- ❑ Public Policies: Not paying own taxes vs keeping other's tax payment
- ❑ Why do people evade?
 - ❑ Absence of guilt or shame: could have latter without former. Being caught the issue rather than any moral concern.
- ❑ Limits to increasing level of penalty: because of danger of wrong conviction
- ❑ Tax amnesties: brings people into net but upsets honest taxpayers. Tax due plus interest: latter let off sometimes

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Further Discussion on Tax Evasion

☐ *Opportunities for Evasion*

- ☐ Self-employed
- ☐ Services in kind: dentist and lawyer friends. Evasion or avoidance?
- ☐ Smuggling and indirect taxes: arises from higher prices across borders

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Can we use a tax policy to solve the problem of negative externalities?

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Please give three examples of
negative externalities.

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A Pigouvian tax

- ☐ **Negative externalities** --- activities that cause harmful effects on other people or the environment and are not reflected in the market price.
- ☐ Example: A factory that pollutes the air with CO₂ may not pay for the damage it causes to the health of nearby residents or the quality of the atmosphere.
- ☐ A Pigouvian tax aims to correct this market failure by making the polluter pay for the social cost of their actions.
 - ☐ CBAM (Carbon Border Adjustment Mechanism --- EU) will take effect in October 2023
 - ☐ Carbon Border Tax is an import tax levied on goods produced by countries that do not have a carbon tax and to support climate change (mostly used in large heavy industries that use intensive energy and have high greenhouse gas emissions.)
 - ☐ Exports of Plastics, steels and aluminums from Thailand will be affected.

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A Pigouvian tax

❑ A Pigouvian tax:

- ❑ Sin tax --- a tax on goods or services that are considered harmful or addictive, such as tobacco, alcohol, or gambling.
- ❑ A sin tax can discourage excessive consumption of these products, which can have negative effects on health, productivity, or social welfare.
- ❑ A sin tax can also help cover the public health costs associated with these products, such as medical treatment or prevention programs.

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A Pigouvian tax

❑ A Pigouvian tax:

- ❑ Congestion pricing --- a fee for driving in a crowded area, such as a city center, during peak hours.
- ❑ A congestion pricing can reduce traffic congestion and pollution, which can improve air quality, road safety, and travel time.
- ❑ A congestion pricing can also encourage drivers to use public transportation or alternative modes of travel, such as cycling or walking.
- ❑ FYI: <https://gothamist.com/news/3-global-cities-have-had-congestion-pricing-for-decades-hows-it-going>

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Tax incidence in environments without perfect competition

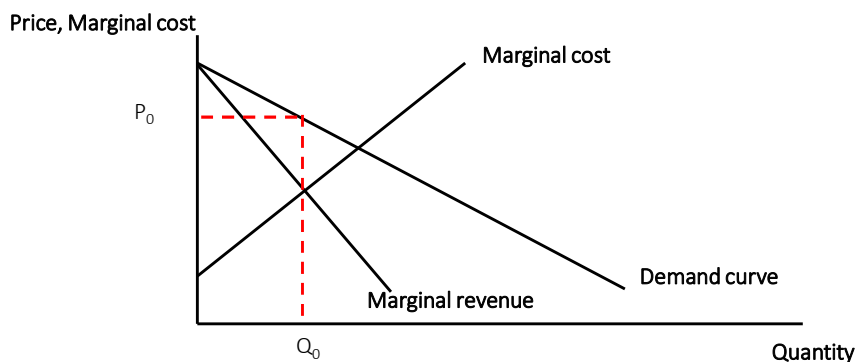
- ❑ **Marginal Revenue (MR)** is the **additional revenue** received by selling **an additional unit** of commodity.
- ❑ **Marginal Cost (MC)** is the **additional cost** when selling/producing **an additional unit** of commodity.
- ❑ In the imperfect competition (a few sellers/monopolist in the market, a product has got very few substitutes), the seller has $MR < P$. (MR and price on the demand curve are not the same.)
 - ❑ e.g. A seller sells 1,000 units of a product at 10 baht per unit, total revenue is $1,000 \times 10 = 10,000$ baht.
 - ❑ The seller wants to sell 2,000 units, he has to lower the price (let's say to 9 baht per unit) and total revenue is now $2,000 \times 9 = 18,000$ baht.
 - ❑ Given that the price is now 9 baht, MR is $(18,000 - 10,000)/(2,000 - 1,000) = 8$ baht per an additional unit of commodity sold. (By lowering the price, his revenue from the previous 1,000 units is also reduced.)

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Tax incidence in environments without perfect competition

- ❑ Optimal condition for a profit maximization:
 - ❑ Choose the quantity level that gives $MR = MC$
 - ❑ The corresponding price will be on the demand curve.



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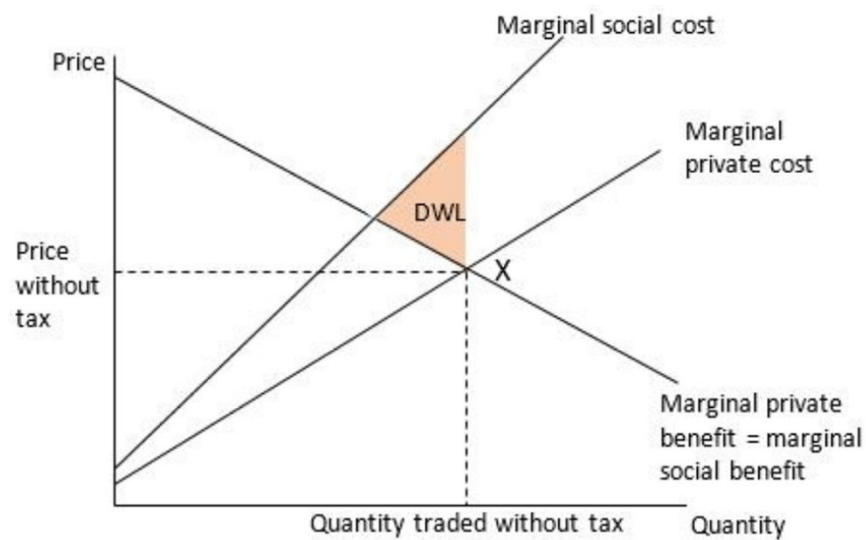
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Tax incidence in environments without perfect competition

❑ A Pigouvian tax --- See “**Good Tax Policy: Taxing Negative Externalities**”

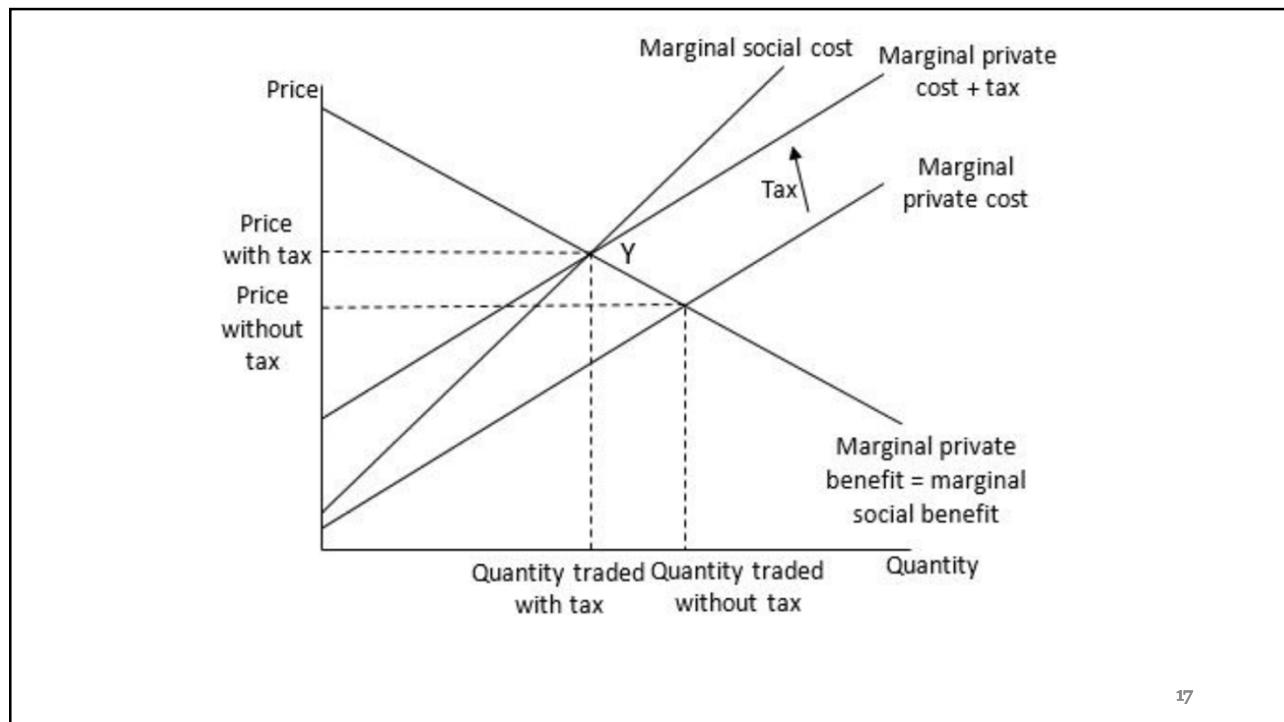
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Conclusion

- ❑ **An externality is a cost passed to bystanders.** For example, the factory sells cloth; the village downstream gets the dirty water. The cost is real, but nobody in the transaction pays it.
- ❑ **Private cost $\neq$ social cost.** Marginal social cost = marginal private cost + marginal external cost ($MSC = MPC + MEC$). The market price reflects only the private half.
- ❑ **As a result, the market overproduces.** This is because the good looks cheaper than it truly is.
- ❑ **Pigou's fix: make the polluter pay the damage.** Set a tax equal to the marginal external cost at the efficient level. The tax shifts supply up by exactly the harm being ignored.
- ❑ **The tax internalises the externality.** Firms now face the full social cost, output falls and pollution falls with it.
- ❑ **But it is hard in practice.** Valuing the damage in a monetary term (dollar, baht, yen), polluters differing in abatement cost, regressive effects on poor households, and rival instruments (regulation, tradable permits, Coasean bargaining) all complicate the clean textbook answer.
- ❑ **An ordinary tax creates a deadweight loss; a Pigouvian tax removes one.**

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